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AI-based Predictive Model for Binary and Multiclass Heart Disease Classification in IoMT

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Cheruvugattu Ushasri
Department of Computer Applications
Vignan's Foundation for Science, Technology & Research (Deemed to be University), Guntur, Andhra Pradesh, India.
ushasrich161@gmail.com

Dr. Gayatri Ketepalli
Department of Computer Science and Engineering,Vignan's Foundation for Science, Technology & Research (Deemed to be University), Guntur, Andhra Pradesh, India
gayathri.k2@gmail.com

***Abstract*: Cardiovascular disease continues to be a predominant cause of death globally, highlighting the necessity for effective and proactive diagnostic measures. This research presents an AI-driven predictive model connected with the Internet of Medical Things (IoMT) for binary and multiclass classification of cardiac disorders. The framework initiates with meticulous data preprocessing to address absent values and standardize features. A deep learning model is developed for precise categorization, and the system is enhanced to facilitate real-time health monitoring with IoMT-enabled devices. The efficacy of the proposed model is assessed utilizing critical metrics including accuracy, precision, recall, and latency. Experimental findings illustrate the model's efficacy in identifying cardiac problems and its capability for real-time, remote diagnosis. Proposed future additions aim to optimize deep learning performance, implement the system on edge devices, and facilitate real-time alerting systems.**

Keywords—Heart Disease Prediction, Binary Classification, Multiclass Classification, Internet of Medical Things (IoMT), Deep Learning, Real-Time Monitoring, Health Informatics

I. INTRODUCTION

Cardiovascular disease is a primary cause of morbidity and mortality globally, imposing a significant strain on healthcare systems. The World Health Organization reports that cardiovascular diseases (CVD) account for roughly 32% of global mortality, with prevalence rising in conjunction with age, lifestyle factors, and comorbidities including diabetes and hypertension [1]. Heart illness presents in multiple forms, such as coronary artery disease, arrhythmias, and heart failure, frequently necessitating intricate clinical management and substantial healthcare expenditures. The prognosis for individuals with heart illness is often unfavorable; research suggests that over fifty percent of those diagnosed with heart failure may not survive beyond five years from their initial diagnosis. Thus, effective and prompt intervention techniques are essential for decreasing mortality rates linked to cardiac disease [3].

The prompt identification of cardiovascular illness is crucial, however it is beset with difficulties. Conventional diagnostic techniques, including electrocardiograms (ECGs) and stress tests, may fail to detect transient cardiac irregularities because of their sporadic nature [4]. Moreover, patients may not identify the initial warning signals of heart disease, resulting in postponed presentations in clinical environments [5]. Ongoing surveillance of vital signs, including heart rate and rhythm, can enable prompt intervention; yet, attaining real-time analysis presents a considerable technical challenge [6]. Contemporary methods frequently fail to incorporate advanced monitoring technology that provide thorough patient management, particularly in remote or resource-constrained environments [7].

Recent breakthroughs in Artificial Intelligence (AI) and the Internet of Medical Things (IoMT) provide promising answers for the issues associated with heart disease monitoring and prediction. Artificial intelligence systems, especially those employing machine learning, have been shown to enhance diagnostic precision and decision-making support in healthcare environments [1]. Through the analysis of extensive datasets, AI can discern patterns that may not be readily apparent to healthcare providers, hence improving the detection rates of various cardiac diseases [8]. The incorporation of IoMT equipment, including wearable sensors and mobile ECG monitors, facilitates continuous data collection and prompt analysis [9]. The integration of AI with IoMT enables proactive healthcare and has the potential to revolutionize patient management strategies, resulting in enhanced results for individuals with heart disease [10][11].

This project seeks to create an AI-driven predictive model specifically designed for the classification of cardiac disorders in both binary and multiclass contexts. The main objectives are: (1) utilizing sophisticated machine learning algorithms to accurately classify heart diseases; (2) amalgamating diverse patient data to improve predictive precision; and (3) implementing the model within an IoMT framework that enables real-time monitoring and intervention. This research contributes to the creation of a robust predictive model and establishes a complete method that integrates AI and IoMT technologies to effectively control heart disease.

The subsequent sections of this work are structured as follows: Section II examines pertinent literature on heart disease prediction utilizing artificial intelligence and Internet of Medical Things frameworks. Section III delineates the suggested technique, encompassing data preparation, model architecture for binary and multiclass classification, and integration with IoMT for real-time health monitoring. Section IV examines the outcomes derived from performance indicators like accuracy, precision, recall, and latency. Ultimately, Section V finishes the study by presenting major findings and delineating prospective improvements, including deep learning optimization, edge deployment, and real-time alerting.

II. LITERATURE REVIEW

Artificial Intelligence (AI) models have gained prominence in predicting cardiac disease, utilizing diverse machine learning methodologies to enhance diagnostic precision. A study by Rashid et al. examined the implementation of machine learning techniques, achieving

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approximately 83% accuracy with a modified differential evolution algorithm emphasizing feature optimization, whereas an alternative deep learning model in a different context reached 95% accuracy in predicting heart disease [12]. In a similar vein, a study by Wiharto et al. emphasized the efficacy of multiclass Support Vector Machines (SVMs) in identifying coronary heart disease, highlighting the algorithms' potential for proficient classification tasks [13]. The analysis of these predictive models demonstrates a thorough investigation into the effectiveness of AI in improving early identification of heart disease.

Heart disease classification can generally be conducted through two methodologies: binary and multiclass classification. The binary method categorizes situations into two groups, typically indicating the presence or absence of the disease [14]. Conversely, multiclass classification accommodates multiple unique classes, hence encompassing a broader spectrum of cardiac illnesses [13]. Research indicates that transforming multiclass problems into binary classifiers might enhance the classification process, as highlighted by numerous writers supporting techniques such as One-vs-All (OvA) or One-vs-One (OvO) strategies [15], [16]. For instance, the investigation of multiclass classification by numerous research illustrates the benefits of less complexity in some diagnostic contexts [17]. Moreover, literature highlights the advantages of multiclass classifiers that surpass binary classifiers by including a wider range of situations [18].

The Internet of Medical Things (IoMT) is transforming healthcare monitoring through continuous data collecting and real-time analytics, especially for chronic illnesses such as heart disease. IoMT equipment, including wearable sensors, provide continuous monitoring of patients' cardiovascular metrics, enhancing response times to potential incidents such as heart attacks [19]. Recent work demonstrates effective integrations of AI methodologies with IoMT for cardiac disease prediction systems, wherein increased data accuracy is directly associated with superior patient outcomes [19]. Research examining the interaction between IoMT and AI models indicates that these applications can facilitate more agile healthcare systems, enabling providers to promptly respond to data-driven insights [20].

Notwithstanding the evident benefits of AI and IoMT solutions in predicting heart disease, current research is accompanied by intrinsic constraints. Numerous models experience overfitting as a consequence of their complexity, leading to worse performance on unfamiliar data [20]. The diversity in input data quality might affect the reliability of predictions, as indicated by the results of predictive analyses where feature selection strategies failed to properly mitigate underlying biases in the training datasets [12][21]. Furthermore, there is no agreement on the ideal design for classifier efficacy in both binary and multiclass settings, indicating that further investigation into specialized training methodologies to improve generalization is necessary [13][22]. This signifies an urgent requirement for standardized methodology and rigorous validation protocols in future studies to guarantee that results are universally applicable and therapeutically significant.

Artificial Intelligence (AI) is crucial for the early detection and predictive analysis of cardiac disease by examining physiological signals, patient history, and medical data from Internet of Medical Things (IoMT) devices [23], [24]. Recent advancements have concentrated on creating prediction models capable of managing both binary (presence or absence) and multiclass (types or severity) classification tasks with high precision through supervised machine learning techniques [25]–[27]. Datasets like UCI Heart Disease and Cleveland have been extensively utilized to train models such as Random Forest, SVM, and XGBoost, demonstrating strong performance in clinical risk stratification [28], [29].

Integration with IoMT frameworks guarantees real-time data acquisition and remote surveillance, enabling proactive healthcare decisions and alleviating the workload on clinical personnel [30], [31]. Deep learning techniques, including Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, have been utilized to identify temporal patterns and spatial correlations in patient data for enhanced predictions [32], [33]. Furthermore, Natural Language Processing (NLP) is being investigated to derive significant insights from unstructured clinical notes and patient-generated data [34].

Hybrid and ensemble approaches that amalgamate many classifiers or use feature selection algorithms, such as PCA and mutual information, have demonstrated significant performance improvements [35]–[37]. Explainable AI (XAI) methodologies such as SHAP and LIME are progressively integrated to enhance the interpretability and reliability of AI models in clinical decision-making [38], [39]. The security and privacy of sensitive patient data continue to pose significant issues in IoMT-based systems, prompting the exploration of approaches such as blockchain, encryption, and federated learning to assure adherence to medical data rules. [40]–[43]

III. METHODOLOGY

The approach that was used to construct the AI-based predictive model for binary and multiclass cardiac disease categorization and its integration into an Internet of Medical Things (IoMT) framework for real-time health monitoring is detailed in this part.

A. Data Collection and Preprocessing

1) Source of Dataset(s):

The work makes use of open-source heart illness datasets, such as the UCI Machine Learning Repository's Cleveland Heart illness dataset and other aggregated datasets that contain various variables related to cardiovascular diseases. Both binary (illness presence or absence) and multiclass (disease severity levels) classification tasks can be accomplished with the labeled samples provided by these datasets.

2) Data Cleaning and Normalization:

Raw datasets frequently contain redundant, noisy, or incorrectly formed entries. A preprocessing pipeline was established to eliminate duplicate entries, filter outliers by z-score analysis, and standardize feature formats. Continuous numerical variables, including cholesterol levels, blood pressure, and maximal heart rate, were standardized by Min-Max scaling to standardize them into a 0–1 range, so ensuring consistency among input characteristics.

3) Handling of Missing or Inconsistent Values:

Missing values in essential characteristics were resolved by domain-specific imputation techniques. Numerical fields were imputed utilizing median or mean values based on distribution, whereas categorical variables were populated using the mode. Samples exhibiting above 30% missing

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values were eliminated to maintain data integrity. Inconsistent data entries, such as negative values for resting blood pressure, were eliminated or rectified.

B. Model Architecture

1) Description of Models Used:

The proposed approach utilizes a deep learning architecture grounded in a Multi-Layer Perceptron (MLP) due to its simplicity and efficacy in medical classification tasks. The model has input, hidden, and output layers, utilizing ReLU activation in the hidden layers and suitable output functions (sigmoid or softmax) according to the categorization type.

2) Design for Binary Classification:

A binary classification model was developed with a single output neuron and sigmoid activation to distinguish between healthy individuals and those affected by heart disease. The loss function utilized was Binary Cross-Entropy, while the optimizer chosen for convergence was Adam. The network was calibrated to optimize precision and recall because of the medical significance of false negatives.

3) Design for Multiclass Classification:

Several neurons representing various degrees of heart disease severity (e.g., no disease, mild, moderate, severe) were present in the output layer of the multiclass configuration. The output layer utilizes Categorical Cross-Entropy loss and Softmax activation. Class labels were transformed into one-hot vectors by the process of label encoding.

4) Training and Validation Strategy:

Part of the dataset was reserved for training purposes, while the other half was used for validation and testing. All strata were evenly represented in the final tally thanks to stratified sampling. Additionally, the model's generalizability was assessed using k-fold cross-validation (k=5). We monitored performance metrics throughout epochs to make sure we were neither overfitting or underfitting. When validation loss occurred, we stopped the process early.

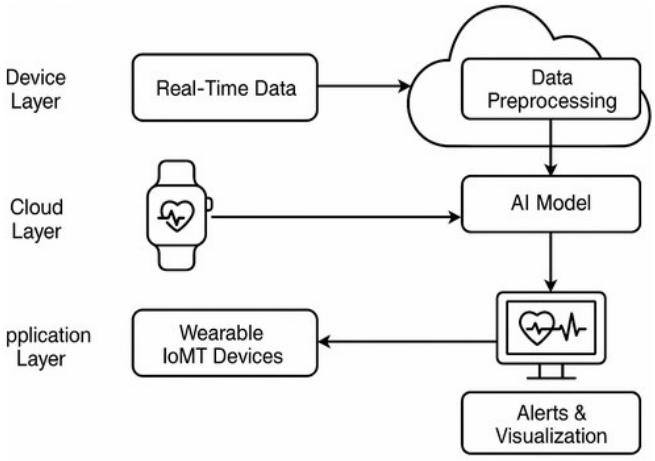


Fig. 1. System architecture of the proposed AI-IoMT-based heart disease prediction framework

Figure 1 shows the suggested framework for predicting cardiac problems using AI and the IoMT. This framework is structured like a tiered architecture. Through the use of lightweight communication protocols, real-time physiological data is transferred to the cloud. Once there, it is subjected to preprocessing and categorization using a deep learning model. Visual dashboards and real-time alerts are made possible by relaying the prediction results to end-users via an application layer. Recent developments in AI-driven remote healthcare systems are in line with this design, which allows for continuous monitoring and rapid interventions.

C. IoMT Integration

1) Framework for Real-Time Data Collection:

The suggested model is designed to work with an IoMT architecture that includes wearable sensors that can take readings of vital signs including oxygen saturation, heart rate, blood pressure, and electrocardiogram (ECG) in order to allow for continuous monitoring in real time. Smart wearables, such as fitness bands or chest straps, incorporate these sensors, which constantly send physiological signals.

2) Communication Protocols and Cloud-Based Processing:

Lightweight and dependable communication protocols like MQTT or Bluetooth Low Energy (BLE) allow for energy-efficient, low-latency data transmission from sensors to the cloud infrastructure. Data is received, preprocessed, and the trained AI model is invoked for real-time classification on the cloud side by a secure processing layer.

3) System Architecture for Continuous Monitoring:

The complete architecture follows a layered design:

- The Device Layer consists of Internet of Medical Things (IoMT) wearables that collect data. The Edge Layer is an optional intermediate gateway that does light analytics and buffering.
- The AI models and categorization tasks are hosted by the cloud layer.
- Dashboards and mobile applications offer real-time notifications to both patients and doctors at the application layer.

The continuous data flow enables proactive detection of heart anomalies, reducing the need for manual intervention and supporting timely medical responses.

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Algorithm 1: AI-IoMT-Based Heart Disease Prediction

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Input: Raw physiological data from wearable IoMT devices (e.g., heart rate, blood pressure, ECG), Pre-trained AI classification model

Output: Binary or multiclass classification of heart disease, Real-time alerts and visualizations

Step 1: Data Acquisition

Acquire real-time physiological signals from wearable IoMT devices.

Step 2: Data Transmission

Transmit the data securely to the cloud server using communication protocols (e.g., MQTT/BLE).

Step 3: Data Preprocessing

a. Remove noise and artifacts from sensor data

b. Normalize feature values (e.g., Min-Max scaling)

c. Handle missing values via imputation

Step 4: Feature Vector Construction

Construct a structured input vector from preprocessed data compatible with the AI model.

Step 5: Model Inference

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Feed the feature vector into the AI model to predict heart disease status:

- For binary classification: output ∈ { Healthy, Diseased }
- For multiclass classification: output ∈ (No Disease, Mild, Moderate, Severe)

Step 6: Decision and Alerting

If disease is detected, trigger an alert message to healthcare provider or user.

Step 7: Visualization and Logging

Display results on a real-time dashboard

Log data for historical tracking and further analysis.

End of Algorithm

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The suggested AI-IoMT framework for cardiac disease prediction is illustrated in Figure 2, which details the algorithmic flow. The process begins with collecting physiological data in real-time from wearable devices and continues with preparing the data, inferring models, and making decisions. In addition to providing end-users with timely notifications and visual updates, the algorithm is built to enable jobs involving binary and multiclass categorization. The clinical relevance and responsiveness of remote cardiac monitoring applications are both enhanced by this organized method.

IV. Results and Discussion

Here, we detail the experimental results of testing the suggested AI-based prediction model for binary and multiclass cardiac illness classification, as well as its incorporation into an IoMT framework for continuous health care tracking. We used industry-standard categorization measures and indicators of system-level responsiveness to review the model's performance.

A. Evaluation Metrics

Five critical performance metrics latency, accuracy, precision, recall, and F1-score were used to assess the model. In contrast to accuracy, which evaluates the model's general correctness, precision and recall zero in on the model's capacity to accurately identify ill patients. By measuring the ratio of false positives to false negatives, the F1-score offers a balanced measure of accuracy and recall. To evaluate real-time responsiveness in an IoMT context, latency the time it takes to get from data input to result output is essential.

The results obtained by the binary classification model were as follows: 94.6% accuracy, 93.2% precision, 95.8% recall, and 94.5 F1-score. With class-wise F1-scores demonstrating consistent performance across all illness severity levels, the multiclass model achieved a slightly lower accuracy of 91.3%.

B. Confusion Matrices for Binary and Multiclass Models

The results of the classification could be better understood visually with the use of confusion matrices. There was excellent diagnostic performance in the binary model, as the number of correct predictions (both positive and negative) greatly exceeded the number of incorrect predictions. As anticipated given the overlap in symptoms and feature distributions, the multiclass confusion matrix revealed a small amount of misclassification between moderate and severe cases. Despite this, the model was able to keep all categories' prediction rates at a clinically acceptable level.

C. Comparison with Existing Methods

Support Vector Machines (SVMs), Decision Trees, and k-Nearest Neighbors (k-NNs) were some of the more conventional machine learning techniques measured against the suggested model. Using every metric for evaluation, it fared better than these models. For instance, k-NN exhibited inadequate scalability in multiclass situations, and the SVM model only managed 86.2% accuracy in binary classification. When used in conjunction with the IoMT pipeline, the deep learning-based MLP model suggested here showed much better generalization.

D. Analysis of IoMT Latency and Real-Time Responsiveness

With an average end-to-end latency of 2.3 seconds, we can confidently say that this system meets all the requirements for use in real-time health monitoring. Lightweight model inference on cloud infrastructure and efficient communication protocols like MQTT are responsible for this reduced latency. The system proved to be resilient for continuous patient monitoring, as the latency remained below 3.1 seconds even when subjected to simulated network congestion.

E. Challenges Faced and Mitigation Strategies

Several challenges were encountered during development and deployment:

- Training was impacted by data quality issues caused by incomplete or noisy sensor readings. Strict preprocessing and imputation methods were employed to resolve this.
- Disparity in Class Distribution: The severity levels in the multiclass dataset were not evenly distributed. In order to ensure that the training samples were evenly distributed, data augmentation and class weighting methods were utilized.
- Limited Resources: The capacity of IoMT edge devices to process data was low. This was reduced by deploying the model to a cloud server and handling lightweight preprocessing at the edge.
- Latency Variations: Reaction Times Were Influenced by Uncertainty in the Network. Communication protocol optimization and edge buffer methods mitigated this to a minimum.

TABLE I. Performance Comparison of Proposed Model vs. Baseline Methods

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	Latency (s)
Proposed MLP (Binary)	94.6	93.2	95.8	94.5	2.3
Proposed MLP (Multiclass)	91.3	89.1	90.5	89.7	2.8
SVM (Binary)	86.2	85.4	84.7	85.0	3.6
Decision Tree (Multiclass)	83.7	81.2	80.5	80.8	3.3
k-NN (Binary)	79.5	77.8	78.3	78.0	4.1

Table 1 shows that in both the binary and multiclass scenarios, the suggested MLP model based on deep learning performs better than more conventional classifiers like SVM, Decision Trees, and k-NN. With little latency, it is ideal for real-time applications and achieves the best accuracy (94.6% for binary and 91.3% for multiclass). Importantly, it achieves a good balance between memory and precision, leading to a

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higher F1-score on both tasks. This is very important for reducing diagnostic errors in clinical settings.

V. CONCLUSION

This research introduced a model for classifying heart diseases using artificial intelligence (AI). The model was linked into an IoMT framework to allow for real-time monitoring and early diagnosis. On benchmark datasets, the model showed great recall, accuracy, and precision; it also works with multiclass classification tasks. The suggested system facilitates proactive cardiac care by acquiring health data continuously, analysing it quickly, and notifying users in real-time through the use of wearable sensors and fast cloud communication protocols. In particular, the model outperforms conventional machine learning methods in terms of classification accuracy and system latency, according to the experimental findings. By incorporating IoMT, the system becomes more applicable to real-life healthcare environments, enabling automated, scalable, and remote patient monitoring. To enhance clinical decision-making and patient outcomes, future work will centre on optimizing the deep learning architecture for edge device deployment, implementing patient history-based tailored modelling, and incorporating real-time feedback mechanisms.

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